Mini-Assessment — Multi-Step Prompt Workflow

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Task

Design a 3-step prompt workflow for processing a student leave request.

The workflow follows: **Step 1: Extract** → **Step 2: Reason** → **Step 3: Format**

- Step 1 uses Few-Shot Prompting
- Step 2 uses Chain-of-Thought reasoning
- Step 3 uses JSON Mode

Objective

The objective is to convert an unstructured student leave request into structured information. The workflow extracts important details, determines the urgency of the request, and produces a valid JSON response.

Input — Student Leave Request

Raw Request:

Hi, I am Yashwanth. I need leave for two days because I have to attend an important family function in Vijayawada. I need to travel tomorrow morning, so please approve my leave request as soon as possible.

Step 1 — Extract Key Entities Using Few-Shot Prompting

Prompt

You are a student information extraction assistant.
Extract the important entities from the student's leave request, such as:
- Student name
- Leave duration
- Reason for leave
- Location
- Travel information
- Deadline
- Student request

Follow the examples below.

Example 1

Leave Request: I am Rahul. I need leave for one day to attend a medical appointment in Hyderabad. I need to leave tomorrow morning.
Extracted Entities:
- Student: Rahul
- Leave Duration: One day
- Reason: Medical appointment
- Location: Hyderabad
- Travel Information: Leaving tomorrow morning
- Deadline: Tomorrow morning
- Request: Leave approval

Example 2

Leave Request: Sneha requests three days of leave to attend her sister's wedding in Chennai. She needs to travel tonight.
Extracted Entities:
- Student: Sneha
- Leave Duration: Three days
- Reason: Sister's wedding

• Location: Chennai
• Travel Information: Travelling tonight
• Deadline: Tonight
• Request: Leave approval

Now process this request: {STUDENT_LEAVE_REQUEST}
Extract the key entities as a clear bullet list.

Step 1 Output

• Student: Yashwanth

• Leave Duration: Two days

• Reason: Important family function

• Location: Vijayawada

• Travel Information: Travelling tomorrow morning

• Deadline: Tomorrow morning

• Student Request: Leave approval

Step 2 — Determine Leave Urgency Using Chain-of-Thought

Prompt

You are a student leave request urgency classifier.
Use the extracted information below to determine the urgency level.

Possible urgency levels:
• Low: The request is not time-sensitive and can be processed later.
• Medium: The request is important but there is no immediate deadline.
• High: The request has an immediate deadline or requires quick action.

Analyze the information step by step by considering:
1. What is the student's main request?
2. What is the reason for the leave?
3. Is there an immediate deadline?
4. Does the student need to travel soon?
5. Based on these factors, determine the urgency level.

Extracted Entities: {STEP_1_OUTPUT}

Provide your reasoning followed by the final urgency level in this format:
Reasoning: [brief explanation]
Urgency: [low/medium/high]

Step 2 Output

Reasoning: The student is requesting two days of leave to attend an important family function in Vijayawada. The student needs to travel tomorrow morning, creating an immediate deadline for leave approval. Therefore, the request requires quick attention.

Urgency: high

Step 3 — Format the Final Result Using JSON Mode

Prompt

You are a data formatting assistant.
Using the extracted entities and urgency classification below, create a valid JSON object.

Requirements:
1. Return only valid JSON.
2. Do not include Markdown formatting.
3. Do not include explanations before or after the JSON.
4. The JSON must contain exactly these three keys: "entities", "urgency", and "summary".
5. "entities" must be an array containing the key information extracted from the request.
6. "urgency" must contain the urgency result from Step 2.
7. "summary" must be a short, clear summary of the leave request.

Step 1 Output: {STEP_1_OUTPUT}
Step 2 Output: {STEP_2_OUTPUT}

Return the final result as valid JSON only.

Step 3 Final Output

```
{
  "entities": [
    "Student: Yashwanth",
    "Leave Duration: Two days",
    "Reason: Important family function",
    "Location: Vijayawada",
    "Deadline: Tomorrow morning",
    "Student Request: Leave approval"
  ],
  "urgency": "high",
  "summary": "Yashwanth requests two days of leave to attend an important family function in Vijayawada and needs approval befo
}
```

End-to-End Workflow

Stage	Action	Output
Input	Student leave request	Yashwanth requests two days of leave for an important family function.
Step 1	Few-Shot Entity Extraction	Student, duration, reason, location, deadline, and request extracted.
Step 2	Chain-of-Thought Urgency Reasoning	Urgency = High
Step 3	JSON Mode Formatting	Valid JSON containing entities, urgency, and summary

Workflow Flow

Student Leave Request → **Step 1: Few-Shot Entity Extraction** → **Step 2: Chain-of-Thought Urgency Classification** → **Step 3: JSON Mode** → **Final JSON Output**

Call 1 — Few-Shot Entity Extraction

```
step1 = prompt_llm(f"""
You are a student information extraction assistant.
Use the examples provided to extract key entities from the leave request.

Example 1:
Leave Request: Rahul needs one day leave for a medical appointment.
Entities:
- Student: Rahul
- Leave Duration: One day
- Reason: Medical appointment

Example 2:
Leave Request: Sneha needs three days leave for her sister's wedding.
Entities:
- Student: Sneha
- Leave Duration: Three days
- Reason: Sister's wedding

Now extract key entities from:
{student_leave_request}

Return a bullet list.
""")
```

Intermediate Output — Step 1

```
- Student: Yashwanth
- Leave Duration: Two days
- Reason: Important family function
- Location: Vijayawada
- Deadline: Tomorrow morning
- Student Request: Leave approval
```

Call 2 — Chain-of-Thought Urgency Reasoning

```
step2 = prompt_llm(f"""
You are a student leave request urgency classifier.
Use the extracted entities below.
{step1}

Reason through the following:
1. Identify the student's main request.
2. Determine whether there is an immediate deadline.
3. Consider the impact if the request is not approved quickly.
4. Classify urgency as low, medium, or high.

Return:
Reasoning: [brief explanation]
Urgency: [low/medium/high]
""")
```

Intermediate Output — Step 2

```
Reasoning: The student needs two days of leave for an important family
function and must travel tomorrow morning. The immediate travel deadline
means the request requires quick attention.
Urgency: high
```

Call 3 — JSON Mode Final Formatting

```
step3 = prompt_llm(
f"""
Using the following outputs:
Entities:
{step1}

Urgency analysis:
{step2}

Create a valid JSON object with exactly these keys:
"entities": array
"urgency": string
"summary": string

Return only valid JSON with no additional text.
""",
response_format="json"
)
```

JSON Validation

The final output can be checked using:

```
import json
result = json.loads(step3)
print(result)
```

Expected result: A valid JSON object containing exactly the keys **entities**, **urgency**, and **summary**.

Conclusion

This workflow correctly demonstrates prompt chaining.

Step 1 — Few-Shot: Extracts entities from the student's leave request.

Step 2 — Chain-of-Thought: Uses the actual output from Step 1 to reason about urgency.

Step 3 — JSON Mode: Uses the outputs of Steps 1 and 2 to produce a structured, valid JSON object.

The final JSON contains the required keys: **entities**, **urgency**, and **summary**.

The workflow is chained because each step uses the previous step's actual output.